Pre-Calculus Concepts

Section 1: Sequences & Partial Sums

1 In an arithmetic sequence, the 4th term is 19 and the 11th term is 54. What is the first term a_1?

a. 4

b. 5

c. 6

d. 7

2 What is the sum of the first 25 terms of the arithmetic series $3 + 7 + 11 + 15 + \ldots$?

a. $1,250$

b. $1,275$

c. $1,300$

d. $1,325$

3 What is the sum of the infinite geometric series $18 - 12 + 8 - \frac{16}{3} + \ldots$?

a. $\frac{54}{5}$

b. $\frac{36}{5}$

c. 10

d. 12

Section 2: Binomial Expansion & Pascal's Triangle

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4 What is the coefficient of x^3y^2 in the binomial expansion of $(2x - y)^5$?

a. 40

b. -40

c. 80

d. -80

5 What is the sum of all coefficients in the expansion of $(3x - 2y)^6$?

a. 1

b. 64

c. 729

d. 0

6 What is the 4th term in the expansion of $(x + 2)^5$ arranged in descending powers of x?

a. $40x^2$

b. $80x^2$

c. $20x^3$

d. $10x^3$

Section 3: Intuitive Limits & Continuity

7 Evaluate the algebraic limit: $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 2x - 3}$.

a. $\frac{3}{2}$

b. $\frac{6}{5}$

c. $\frac{3}{4}$

d. Does not exist

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8 Evaluate the limit at infinity: $\lim_{x \to \infty} \frac{4x^3 - 5x + 2}{2x^3 + 7x^2 - 1}$.

a. 0

b. 2

c. 4

d. ∞

9 Evaluate the limit involving radicals: $\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. 1

d. 0

Section 4: Difference Quotients & Tangents

10 For $f(x) = 3x^2 - 5x$, simplify the difference quotient $\frac{f(x + h) - f(x)}{h}$ for $h \neq 0$.

a. $6x + 3h - 5$

b. $6x - 5$

c. $6x + 3h$

d. $3x + 3h - 5$

11 What is the slope of the tangent line to the curve $y = x^2 - 4x$ at $x = 3$?

a. 2

b. 3

c. 4

d. 6

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12 At what value of x does the function $f(x) = -2x^2 + 12x - 5$ achieve its maximum value?

a. 2

b. 3

c. 4

d. 6

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